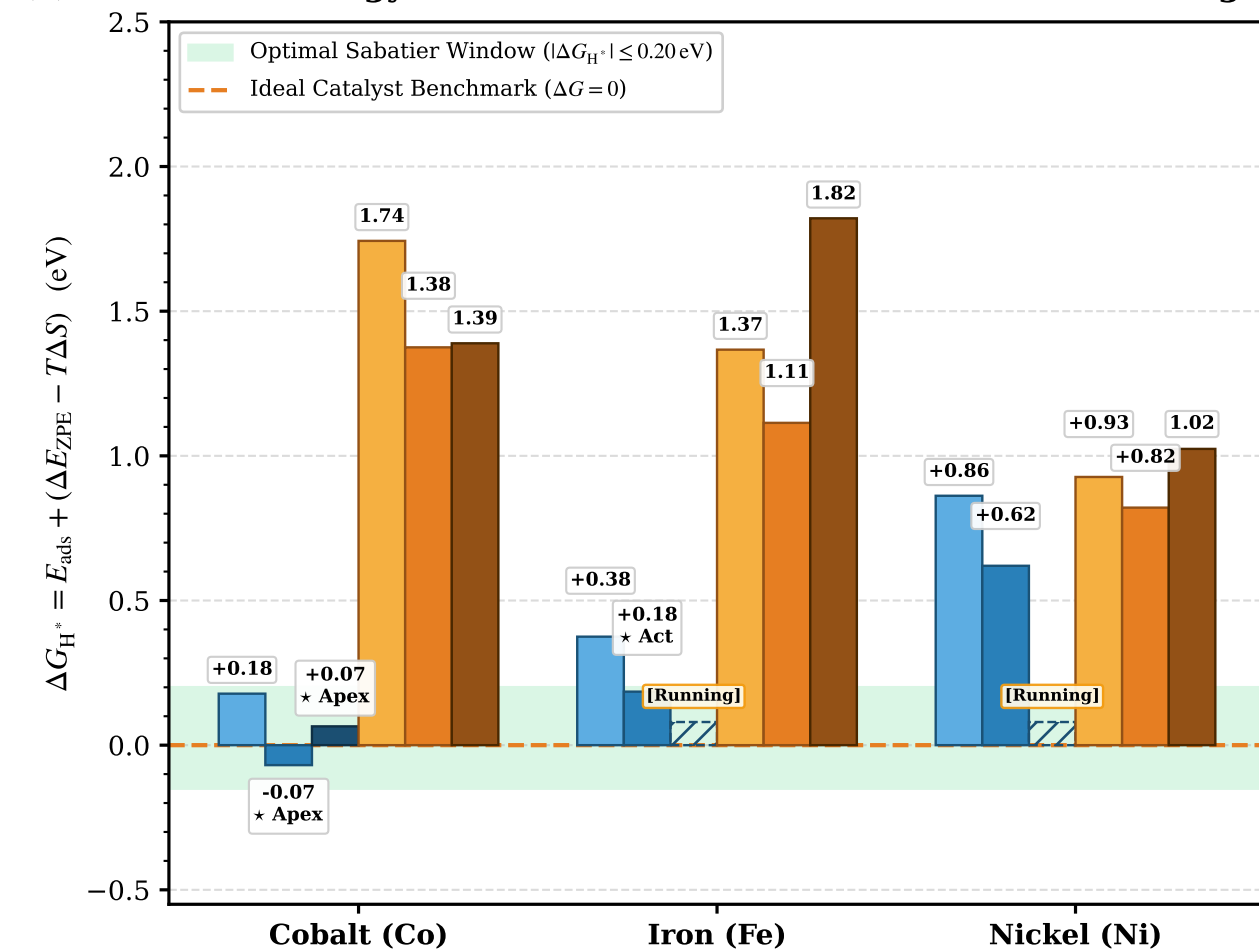


Comprehensive Multitier Benchmark: Adsorbed vs. Embedded CrCl₃-TM (TM = Co, Fe, Ni)

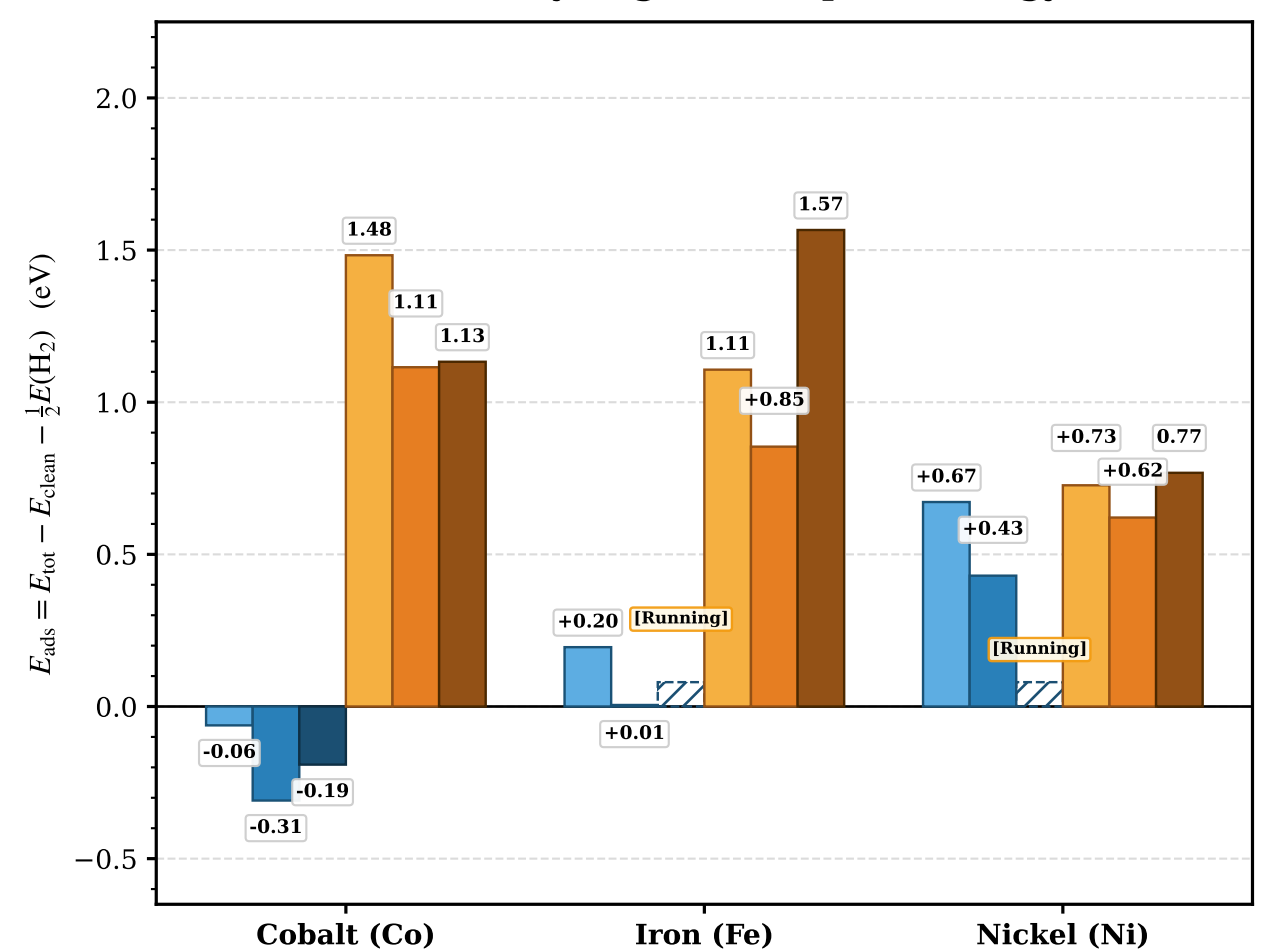
Systematic Comparison across vdW Dispersion, Single-Site + U_{Cr} , and Multi-Site + U_{all} ($U_{\text{Cr}} + U_{\text{TM}}$)

■ Ads: PBE+D3 (vdW)
 ■ Ads: + U_{Cr} (3.29 eV)
 ■ Ads: + U_{all} ($U_{\text{Cr}} + U_{\text{TM}}$)
 ■ Emb: PBE+D3 (vdW)
 ■ Emb: + U_{Cr} (3.29 eV)
 ■ Emb: + U_{all} ($U_{\text{Cr}} + U_{\text{TM}}$)

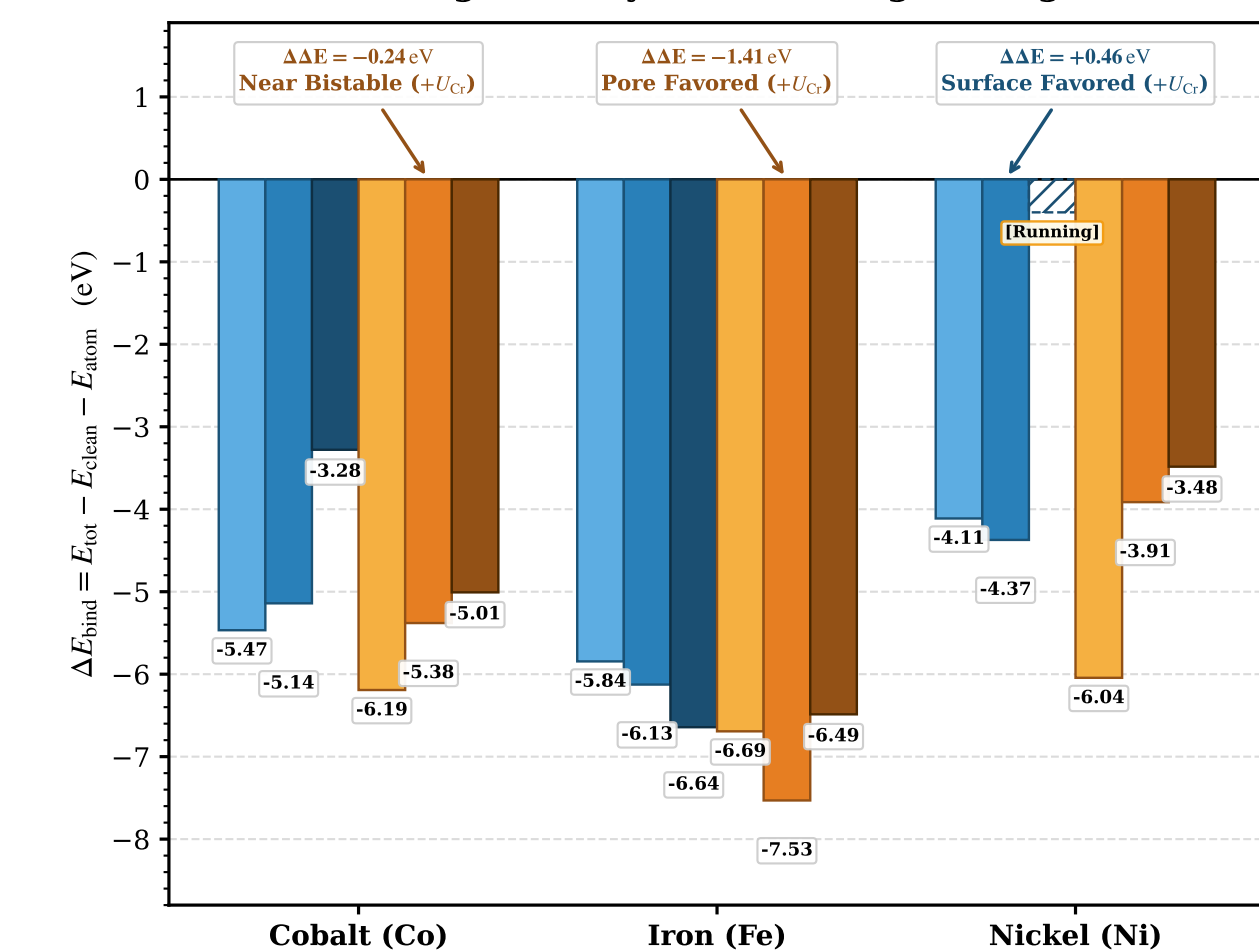
(a) HER Free Energy: Adsorbed vs. Embedded across Methodological Tiers



(b) Electronic Hydrogen Adsorption Energy (E_{ads})



(c) TM Anchoring Stability & Embedding Driving Force ($\Delta\Delta E$)



(d) Spin Polarization & Magnetic Moment Trends (2×2 Supercell)

